

gilmore-lie-groups: residuals

equations measured 2026-09-15T14:55:46Z against the model of 2026-09-15

Corrected (6)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ E00012 (was)	20	■ 0.416	$\begin{array}{ll} (12)(23)=(321) \ \& \ (23)(12)=(123) \ \& \\ (321) \end{array}$	$\begin{array}{l} (12)(23) = (321) \\ (23)(12) = (123) \end{array} \quad (321)$	$\begin{array}{l} (12)(23) = (321) \\ (23)(12) = (123) \end{array} \quad (123) \neq (321)$
gilmore-lie-groups_ E00012 (now)	20	—	$\begin{array}{ll} (12)(23)=(321) \ \& \ (23)(12) \\ \cancel{=(123)} \ \& \ (321) \end{array}$	$\begin{array}{l} (12)(23) = (321) \\ (23)(12) \neq (123) \end{array} \quad (321)$	$\begin{array}{l} (12)(23) = (321) \\ (23)(12) = (123) \end{array} \quad (123) \neq (321)$
basis: census / census					
gilmore-lie-groups_ E00412 (was)	126	■ 1.000	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$
gilmore-lie-groups_ E00412 (now)	126	—	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$	$e^{-i \theta_z \prime / 2} = \cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)$
basis: census / census					
gilmore-lie-groups_ E00413 (was)	126	■ 1.000	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$
gilmore-lie-groups_ E00413 (now)	126	—	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$	$\left\langle \begin{array}{c} j \\ -j \end{array} \middle e^{i \theta \cdot \mathbf{J}} \middle \begin{array}{c} j \\ -j \end{array} \right\rangle = e^{-i j \theta_z \prime} = \left(e^{-i \theta_z \prime / 2} \right)^{2 j} = [\cos (\theta / 2) - i \left(\theta_z / \theta \right) \sin (\theta / 2)]^{2 j}$
basis: census / census					
gilmore-lie-groups_ E00486 (was)	149	■ 0.996	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text {compact subalgebra }} + \underbrace{V_{+}^{\prime}}_{\text {noncompact generators }}}_{\text {nonsemisimple Lie algebra }} + \underbrace{V_0^{\prime}}_{\text {maximum solvable invariant subalgebra }}}_{\text {abelian }}}_{\text {semisimple }}$	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text {compact subalgebra }} + \underbrace{V_{+}^{\prime}}_{\text {noncompact generators }}}_{\text {nonsemisimple Lie algebra }} + \underbrace{V_0^{\prime}}_{\text {maximum solvable invariant subalgebra }}}_{\text {abelian }} \quad \text{semisimple}$	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text {compact subalgebra }} + \underbrace{V_{+}^{\prime}}_{\text {noncompact generators }}}_{\text {semisimple }} + \underbrace{\underbrace{V_0^{\prime}}_{\text {abelian }} + \underbrace{V_0}_{\text {nilpotent }}}_{\text {maximum solvable invariant subalgebra }}}_{\text {nonsemisimple Lie algebra }}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
gilmore-lie-groups_ E0486 (now)	149	—	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text{$\cancel{c}$ompact subalgebra}} + \underbrace{V_{+}^{\prime}}_{\text{noncompact generators}}}_{\text{nonsemisimple Lie $\cancel{a}$lgebra}}} + \underbrace{V_0^{\prime}}_{\text{abelian}} + \underbrace{V_0}_{\text{nilpotent}}$	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text{compact subalgebra}} + \underbrace{V_{+}^{\prime}}_{\text{noncompact generators}}}_{\text{nonsemisimple Lie algebra}} + \underbrace{V_0^{\prime}}_{\text{maximum solvable invariant subalgebra}} + \underbrace{V_0}_{\text{abelian}}}_{\text{semisimple}}$	$\mathfrak{g} = \underbrace{\underbrace{\underbrace{V_{-}^{\prime}}_{\text{compact subalgebra}} + \underbrace{V_{+}^{\prime}}_{\text{noncompact generators}}}_{\text{semisimple}}} + \underbrace{\underbrace{V_0^{\prime}}_{\text{abelian}} + \underbrace{V_0}_{\text{nilpotent}}}_{\text{maximum solvable invariant subalgebra}}_{\text{nonsemisimple Lie algebra}}$
basis: census / census					
gilmore-lie-groups_ E0556 (was)	174	■ 0.447	$\begin{aligned} \{[H_i, H_j], [H_i, H_j]\} &= 0 \\ \{[H, E_\alpha], [H, E_\alpha]\} &= \alpha E_\alpha \\ \{[E_\alpha, E_\beta], [E_\alpha, E_\beta]\} &= \alpha \cdot H \quad \alpha + \beta = 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \end{aligned}$	$\begin{aligned} [H_i, H_j] &= 0 \\ [H, E_\alpha] &= \alpha E_\alpha \\ [E_\alpha, E_\beta] &= \alpha \cdot H \quad \alpha + \beta = 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \end{aligned}$	$\begin{aligned} [H_i, H_j] &= 0 \\ [H, E_\alpha] &= \alpha E_\alpha \\ [E_\alpha, E_\beta] &= \alpha \cdot H \quad \alpha + \beta = 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \quad \alpha + \beta \neq 0 \text{ a root} \\ &= 0 \quad \alpha + \beta \text{ not a root} \end{aligned}$
gilmore-lie-groups_ E0556 (now)	174	—	$\begin{aligned} \{[H_i, H_j], [H_i, H_j]\} &= 0 \\ \{[H, E_\alpha], [H, E_\alpha]\} &= \alpha E_\alpha \\ \{[E_\alpha, E_\beta], [E_\alpha, E_\beta]\} &= \alpha \cdot H \quad \alpha + \beta \neq 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \end{aligned}$	$\begin{aligned} [H_i, H_j] &= 0 \\ [H, E_\alpha] &= \alpha E_\alpha \\ [E_\alpha, E_\beta] &= \alpha \cdot H \quad \alpha + \beta \neq 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \end{aligned}$	$\begin{aligned} [H_i, H_j] &= 0 \\ [H, E_\alpha] &= \alpha E_\alpha \\ [E_\alpha, E_\beta] &= \alpha \cdot H \quad \alpha + \beta = 0 \\ &= N_{\alpha, \beta} E_{\alpha + \beta} \quad \alpha + \beta \neq 0 \text{ a root} \\ &= 0 \quad \alpha + \beta \text{ not a root} \end{aligned}$
basis: census / census					
gilmore-lie-groups_ E0788 (was)	247	■ 0.732	$\begin{aligned} M_{+}^{\prime} Y_m^l &= \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ &\quad - \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \end{aligned}$	$M_{+}^{\prime} Y_m^l = \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ - \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \end{aligned}$	$M_{+}^{\prime} Y_m^l = \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a + 1 & m_b \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ - \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b + 1 \end{array} \right) \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \right) \left \begin{array}{c} l \\ m \end{array} \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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gilmore-lie-groups_ E00788 (now)	247	—	$\begin{aligned} M_{+}^{\prime\prime} Y_m^l = & \left\langle \theta \phi \middle \left(\begin{array}{cc} j/2 & j/2 \\ m_a+1 & m_b \end{array} \right) \begin{array}{c} j/2 \\ m_a \end{array} \begin{array}{c} j/2 \\ m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \\ & - \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b+1 \end{array} \right) \begin{array}{c} j/2 \\ m_a \end{array} \begin{array}{c} j/2 \\ m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \end{array} \rangle$	$M_{+}^{\prime} Y_m^l = \left\langle \theta \phi \middle \left(\begin{array}{cc} j/2 & j/2 \\ m_a+1 & m_b \end{array} \right) \begin{array}{c} j/2 \\ m_a \end{array} \begin{array}{c} j/2 \\ m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)} \right. \\ \left. - \left(\begin{array}{cc} j/2 & j/2 \\ m_a & m_b+1 \end{array} \right) \begin{array}{c} j/2 \\ m_a \end{array} \begin{array}{c} j/2 \\ m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \right)$	$M_{+}^{\prime} Y_m^l = \langle \theta \phi \left(\begin{array}{cc} j/2 & j/2 \\ m_a+1 & m_b \end{array} \right) \begin{array}{c} j/2 \\ m_a \end{array} \begin{array}{c} j/2 \\ m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_a)(j/2 + m_a + 1)}$ $- \left \begin{array}{cc} j/2 & j/2 \\ m_a & m_b+1 \end{array} \right\rangle \left\langle \begin{array}{cc} j/2 & j/2 \\ m_a & m_b \end{array} \middle m \right\rangle \times \sqrt{(j/2 - m_b)(j/2 + m_b + 1)} \Bigg)$
basis: census / census					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E08584 (11.4) [conf 0.001]	188	■ 0.001 ● C +67	\left.\left.\left[\begin{array}{cccccccc} 1 & & & & & & & \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & \ddots & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & -1 & \\ & & & & & & & 1 & \\ & & & & & & & & -1 & \\ & & & & & & & & & \ddots \end{array}\right]\right.\right. \begin{array}{c} H_{1} \\ H_{3} \\ \vdots \\ H_n \\ \frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha}) \\ \frac{1}{\sqrt{2}}(E_{\beta} + E_{-\beta}) \\ \frac{1}{\sqrt{2}}(E_{\beta} - E_{-\beta}) \\ \vdots \end{array}	(not rendered)	1 1 1 ⋱ 1 1 −1 1 −1 ⋱ H_1 H_2 H_3 $\vdots$ H_n $\frac{1}{\sqrt{2}}(E_{\alpha} + E_{-\alpha})$ $\frac{1}{\sqrt{2}}(E_{\alpha} - E_{-\alpha})$ $\frac{1}{\sqrt{2}}(E_{\beta} + E_{-\beta})$ $\frac{1}{\sqrt{2}}(E_{\beta} - E_{-\beta})$ $\vdots$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Flagged, not acted on

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ EQ0624 (1.20)	24	■ 0.222 ● C +21	$\left.\left.\left(r_1-r_2\right)\right.\right.\xrightarrow{I}\left.\left(r_1-r_2\right),$	$(r_1-r_2)\}\xrightarrow{I}+(r_1-r_2),$	$(r_1-r_2)\left\{\begin{array}{l}\xrightarrow{I}+(r_1-r_2)\\ \xrightarrow{(12)}(r_2-r_1)=-(r_1-r_2)\end{array}\right.$
gilmore-lie-groups_ EQ0474 (8.6)	145	■ 0.233 ● C +114	$\begin{gathered}\eta\left(\hat{n}+\frac{1}{2}\right)+Ra^{\dagger\,2}+La^2+ra^{\dagger}+la+\delta I\\ \mathfrak{d}\mathfrak{e}\mathfrak{f}=\begin{bmatrix}0&l&r&-2\delta\\0&\eta&2R&-r\\0&-2L&-\eta&l\\0&0&0&0\end{bmatrix}\\\mathfrak{R}\mathfrak{e}\mathfrak{g}=\left[\begin{array}{ccc ccc}\begin{array}{ccc}0&-2R&2L\\4L&2\eta&0\\-4R&0&-2\eta\end{array}&\begin{array}{ccc}-r&l&0\\2l&0&0\\0&-2r&0\end{array}&\begin{array}{c}\hat{n}+\frac{1}{2}\\a^{\dagger 2}\\a^2\\a^{\dagger}\\a\\I\end{array}\end{array}\right]\end{gathered}$	(not rendered)	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — basis names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ E08368 (7.6) [conf 0.032]	114	0.032	$\operatorname{tr} e^{SXS^{-1}} \longrightarrow \left\{ \begin{array}{ll} 2 \cosh \theta & \geq 2 \\ 2 \cos \theta & \geq -2 \end{array} \right. \begin{array}{l} \text{real eigenvalues} \\ \text{imaginary eigenvalues} \end{array}$	$\operatorname{tr} e^{SXS^{-1}} \longrightarrow \left\{ \begin{array}{ll} 2 \cosh \theta & \geq 2 \\ 2 \cos \theta & \geq -2 \end{array} \right. \begin{array}{l} \text{real eigenvalues} \\ \text{imaginary eigenvalues} \end{array}$	
Conf. MathPix confidence: ≥0.9 0.5-0.9 <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): C component W weak S stable N noise K clean not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					